

Supplementary Material

Supplementary Material

Pretrained Protein Representations Unlock Multimodal Fusion Gains for Enzyme Kinetic Prediction

[Authors anonymized for review]

S1: Frozen Split Identities

- Source SHA256:
13643b0b36374f8d3f64d8b014882cf1b3b58946eeaae2b9dcd59e8b2c2d6719
- Split SHA256: ae1793afea078b4a2d8b834e9b9cd7c836d43508e26185ed87d26b8b82f96300
- MMseqs2 version: 5d152c612b6ad2a56f657b7a02c127ecea2a75
- Clustering: 30% identity, 80% coverage, coverage mode 0
- Clusters: 2,204 (13,157 train / 1,640 validation / 1,646 test after median aggregation)

S2: ESM-2 Model Configuration

- Model: esm2_t6_8M_UR50D (6 layers, 320-dim, 8M parameters, frozen)
- Projection: Linear(320, 256) + SiLU + Dropout(0.1) + LayerNorm(256)
- Chunk pooling: 128-residue chunks, masked mean
- Sequence truncation: 512 residues maximum
- Training: AdamW, lr=3e-4, cosine schedule, 2-epoch warmup, patience 4
- Effective batch size: 32 (micro-batch 4, accumulation 8)
- Model parameters (trainable): 12,946,716

S3: Architecture Ablation Results (Seed 42, From-Scratch Model)

Supplementary Table S1. Single-seed ablation results.

Ablation	Test RMSE	Delta RMSE	Test Pearson
Task-query baseline	1.4638	—	0.2731
Remove reaction context	1.5046	+0.0409	0.1908
Shared task query	1.4911	+0.0273	0.2520
Global-mean protein encoder	1.4910	+0.0272	0.2278

S4: Per-Seed Test Results (ESM-2 Multimodal)

Supplementary Table S2. Per-seed test metrics.

Seed	Test RMSE	Test MAE	Test Pearson	Best Epoch
7	1.3880	1.0683	0.3906	2
42	1.3754	1.0606	0.3968	1

Seed	Test RMSE	Test MAE	Test Pearson	Best Epoch
123	1.4114	1.0835	0.3556	3
Mean	1.3916	1.0708	0.3810	—
SD	0.0183	0.0117	0.0218	—

S5: EnzEngDB Selection Protocol

- Archive: 30,050,527 bytes, SHA256
8013ad81586db2187162aada0709c1cabc7e7d69f03dd5c199776aaf000dd6ea
 - Accepted rows: 245,945 across 160 campaigns
 - Label-independent cap: 2,000 rows per campaign by SHA256 of candidate ID
 - Homology filtering: MMseqs2 30%/80% against all UniKP sequences
 - Final selection: 6,423 rows, 51 campaigns, zero exact sequence overlap
 - Selection SHA256:
8e76932fc76b884ea8cf127934683b43b9e896b057b817488a9dc5533cf1fc4b
-

S6: Claim Boundary

This work establishes an internal homology-cold predictive gain. It does not establish: calibrated uncertainty, external kcat validation, activity or flux prediction, candidate ranking utility, or publication-grade efficacy for enzyme engineering decisions. The prospective independent benchmark remains incomplete (192/300 records, 25/30 families). No model predictions have been generated for the prospective pool.